DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)

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INSTITUTION'S INNOVATION COUNCIL (Ministry of Education Initiative)

1.ABSTRACT

ReSector is a local-first academic research workstation that turns static PDF libraries into an interactive knowledge ecosystem. It combines agentic retrieval-enhanced generation (RAG) with multi-document chat, to receive a citation graph built from open scholarly indexes, the three analysis tools for methodology extraction, plain language translation and adversary peer review. Papers and embeddings stay on the user's machine while the system works with state-of-the-art LLMs and open academic knowledge graphs.

2.PROBLEM STATEMENT

- Literature is growing in width and researchers can read it.
- PDFs are static: no cross-paper search, no ground ansvers, no citation context.
- Generic chatbots hallucinate citations and cannot see a researcher's own papers.
- Understanding a field's structure needs manual citation tracing across databases.
- Cloud-only tools raise data-privacy concerns for unpublished work.

3.PROPOSED SOLUTION

RESECTOR WORKSPACE

- **Chat with Paper** — Multi-PDF grounded Q&A with online citations.
- **Reviewer** — Five reviewer personas from Supportive Peer for Bntal Reviewer 2.
- **Methodology Sifter** — Extracts research questions, method and sample size, key findings, fatal flaws.
- **Citation Graph** — 50+ paper network with Luanah clusters.
- **Jargon Simplifier** — plain-English breakdown plus real-world analogy.

ReSector

An AI-powered, local-first research workstation for intelligent literature exploration, evidence synthesis and scientific discovery

5.KEY FEATURES

Simplified Research for all

50+
Paper per Citation Graph

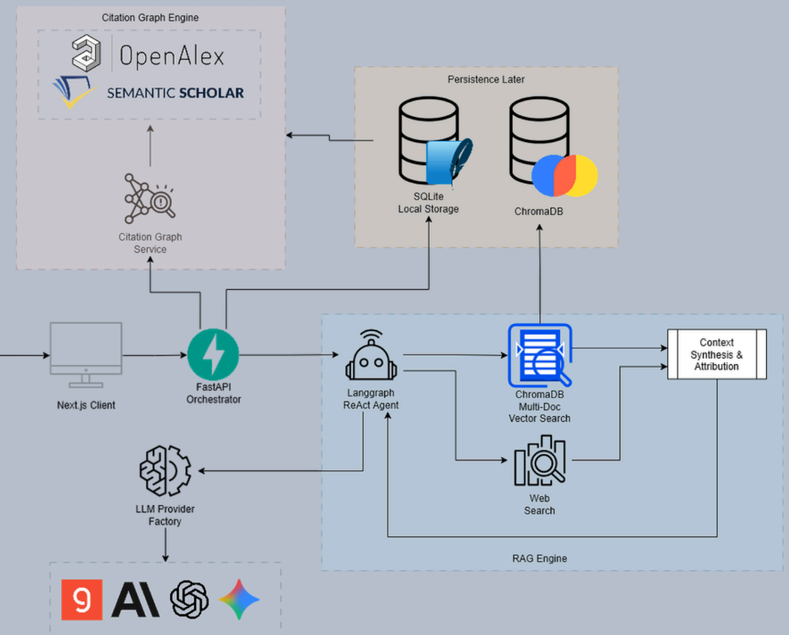
800+
Interconnected Citation Links

4- 8
Thematic Clusters

100%
Local Storage

Try ReSector

4.ARCHITECTURE DIAGRAM



Frontend

Next.js | React 19 | TypeScript | Tailwind CSS | react-force graph-2d

Backend

Python | FastAPI | LangGraph | LangChain | SQLAlchemy | Docker

Data and retrieval

ChromaDB | SQLite | PyMuPDF | MiniLM-L6-v2 embeddings

6.PROGRAM ALIGNMENT

Program outcomes (POs) — filled = mapped

P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011
■	■	■	■	■	■	■		■		■

Program specific outcomes (PSOs)

PSO 1 | PSO 2 | PSO 3

Program education objectives (PEOs)

PEO 1 | PEO 2 | PEO 3 | PEO 4

UN Sustainable Development Goals

- 4 Quality Education
- 9 Industry, Innovation & Infrastructure
- 10 Reduced Inequalities

TRL Level

6

Prototype demonstrated in a relevant environment

7.HOW RESECTOR IS DIFFERENT

vs. NotebookLM: NotebookLM answers from the sources you upload, inside Google's cloud. ReSector keeps your papers and embeddings on your own machine. You choose the model (Groq, OpenAI, Anthropic or Gemini), and it goes beyond Q&A with a citation graph, a methodology sifter and a persona-based peer reviewer.
vs. Semantic Scholar: Semantic Scholar helps you find published papers. ReSector starts from the papers you already have. It chats with them, stress-tests their arguments, and uses Semantic Scholar and OpenAlex as data sources for the citation graph.
In one line: Others help you find or read papers. ReSector helps you understand, question and connect them, privately.

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